

**LAPORAN PRAKTIKUM  
APLIKASI BERBASIS PLATFORM**

**MODUL - 4, 5  
ANTAR MUKA PENGGUNA**



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## TUGAS

Tugas praktikum modul 4-5 Flutter membuat 1 project flutter yang menampilkan beberapa widget UI berikut:

- Yang harus ada:

- Container → kotak berwarna
- GridView → minimal 6 item (grid)
- ListView → 3 item (A, B, C)
- ListView.builder → list dari data array
- ListView.separated → list + garis pembatas
- Stack → tampilan bertumpuk (kotak / text)

 Output yang dikumpulkan: Screenshot hasilnya, Source code, Penjelasan singkat tiap widget.

## 2. Source Code main.dart

```
import 'package:flutter/material.dart';

void main() {
  runApp(const SmartWalletApp());
}

class SmartWalletApp extends StatelessWidget {
  const SmartWalletApp({super.key});
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      title: 'NeoBank Dashboard',
      debugShowCheckedModeBanner: false,
      theme: ThemeData(
        useMaterial3: true,
        colorScheme: ColorScheme.fromSeed(
          seedColor: const Color(0xFF6366F1),
          brightness: Brightness.light,
        ),
      ),
      home: const DashboardScreen(),
    );
  }
}

class DashboardScreen extends StatelessWidget {
  const DashboardScreen({super.key});
  // Data Array untuk ListView.builder
  final List<Map<String, dynamic>> transactions = const [
    {"title": "Apple Music Subscription", "amount": "-\$9.99", "date":
    "Today, 10:24", "icon": Icons.music_note},
    {"title": "Salary Payment", "amount": "+\$4,500.00", "date":
    "Yesterday, 09:00", "icon": Icons.account_balance_wallet},
    {"title": "Starbucks Coffee", "amount": "-\$5.50", "date":
    "Yesterday, 15:30", "icon": Icons.coffee},
```



```

        offset: const Offset(0, 10),
      ),
    ],
  ),
),
Positioned(
  top: 25,
  left: 25,
  child: Column(
    crossAxisAlignment: CrossAxisAlignment.start,
    children: [
      Text("Balance", style: TextStyle(color:
Colors.white.withOpacity(0.8), fontSize: 16)),
      const Text("\$12,450.00", style:
TextStyle(color: Colors.white, fontSize: 32, fontWeight:
FontWeight.bold)),
    ],
  ),
),
const Positioned(
  bottom: 25,
  left: 25,
  child: Text("**** * 4242", style:
TextStyle(color: Colors.white, fontSize: 18, letterSpacing: 2)),
),
Positioned(
  top: 25,
  right: 25,
  child: Container(
    padding: const EdgeInsets.all(8),
    decoration: BoxDecoration(color:
Colors.white.withOpacity(0.2), shape: BoxShape.circle),
    child: const Icon(Icons.contactless, color:
Colors.white),
  ),
),
],
),
const SizedBox(height: 30),
// 2. GRIDVIEW (Quick Actions)
const SectionHeader(title: "Quick Actions", subtitle:
"GridView (6 Items)"),
GridView.count(
  shrinkWrap: true,
  physics: const NeverScrollableScrollPhysics(),
  crossAxisCount: 3,
  mainAxisSpacing: 15,

```

```

        crossAxisSpacing: 15,
        children: [
            _buildMenuIcon(Icons.send, "Send", Colors.blue),
            _buildMenuIcon(Icons.qr_code_scanner, "Scan",
Colors.orange),
            _buildMenuIcon(Icons.receipt_long, "Bills",
Colors.green),
            _buildMenuIcon(Icons.trending_up, "Invest",
Colors.purple),
            _buildMenuIcon(Icons.security, "Insurance",
Colors.red),
            _buildMenuIcon(Icons.more_horiz, "More", Colors.grey),
        ],
    ),
    const SizedBox(height: 30),
    // 3. LISTVIEW (Statis - Account Tiers)
    const SectionHeader(title: "Account Tiers", subtitle:
"ListView Static (A, B, C)"),
    SizedBox(
        height: 100,
        child: ListView(
            scrollDirection: Axis.horizontal,
            children: [
                _buildTierCard("Tier A", "Silver Member",
Colors.blueGrey),
                _buildTierCard("Tier B", "Gold Member",
Colors.orangeAccent),
                _buildTierCard("Tier C", "Platinum Elite",
Colors.indigo),
            ],
        ),
    ),
    const SizedBox(height: 30),
    // 4. LISTVIEW.BUILDER (Transactions)
    const SectionHeader(title: "Recent Transactions",
subtitle: "ListView.builder"),
    ListView.builder(
        shrinkWrap: true,
        physics: const NeverScrollableScrollPhysics(),
        itemCount: transactions.length,
        itemBuilder: (context, index) {
            final item = transactions[index];
            return Container(
                margin: const EdgeInsets.only(bottom: 12),
                padding: const EdgeInsets.all(12),
                decoration: BoxDecoration(color: Colors.white,
borderRadius: BorderRadius.circular(16)),
                child: ListTile(

```

```

        leading: CircleAvatar(
          backgroundColor: const Color(0xFFF1F5F9),
          child: Icon(item['icon'], color: const
Color(0xFF6366F1)),
        ),
        title: Text(item['title'], style: const
TextStyle(fontWeight: FontWeight.bold)),
        subtitle: Text(item['date']),
        trailing: Text(item['amount'], style: TextStyle(
          color: item['amount'].contains('+') ?
Colors.green : Colors.black,
          fontWeight: FontWeight.bold,
          fontSize: 16
        )),
      ),
    );
  },
),
const SizedBox(height: 30),
// 5. LISTVIEW.SEPARATED (Financial Goals)
const SectionHeader(title: "Financial Goals", subtitle:
"ListView.separated"),
Container(
  decoration: BoxDecoration(color: Colors.white,
borderRadius: BorderRadius.circular(16)),
  child: ListView.separated(
    shrinkWrap: true,
    physics: const NeverScrollableScrollPhysics(),
    itemCount: financialGoals.length,
    separatorBuilder: (context, index) => const
Divider(indent: 20, endIndent: 20, height: 1),
    itemBuilder: (context, index) {
      return ListTile(
        leading: const Icon(Icons.flag_circle, color:
Colors.amber),
        title: Text(financialGoals[index]),
      );
    },
  ),
),
const SizedBox(height: 50),
],
),
),
),
);
}

Widget _buildMenuIcon(IconData icon, String label, Color color) {

```

```

    return Column(
      children: [
        Container(
          padding: const EdgeInsets.all(16),
          decoration: BoxDecoration(color: color.withOpacity(0.1),
borderRadius: BorderRadius.circular(20)),
          child: Icon(icon, color: color, size: 28),
        ),
        const SizedBox(height: 8),
        Text(label, style: const TextStyle(fontWeight: FontWeight.w600,
fontSize: 12)),
      ],
    );
  }

  Widget _buildTierCard(String title, String desc, Color color) {
    return Container(
      width: 200,
      margin: const EdgeInsets.only(right: 15),
      padding: const EdgeInsets.all(15),
      decoration: BoxDecoration(color: color, borderRadius:
BorderRadius.circular(20)),
      child: Column(
        crossAxisAlignment: CrossAxisAlignment.start,
        children: [
          Text(title, style: const TextStyle(color: Colors.white,
fontWeight: FontWeight.bold, fontSize: 18)),
          Text(desc, style: TextStyle(color:
Colors.white.withOpacity(0.8))),
        ],
      ),
    );
  }
}

class SectionHeader extends StatelessWidget {
  final String title;
  final String subtitle;
  const SectionHeader({super.key, required this.title, required
this.subtitle});
  @override
  Widget build(BuildContext context) {
    return Padding(
      padding: const EdgeInsets.only(bottom: 15),
      child: Row(
        mainAxisAlignment: MainAxisAlignment.spaceBetween,
        children: [
          Text(title, style: const TextStyle(fontSize: 18, fontWeight:
FontWeight.bold)),

```

```

        Text(subtitle, style: TextStyle(fontSize: 12, color:
Colors.indigo.shade400, fontWeight: FontWeight.w600)),
      ],
    ),
  );
}
}

```

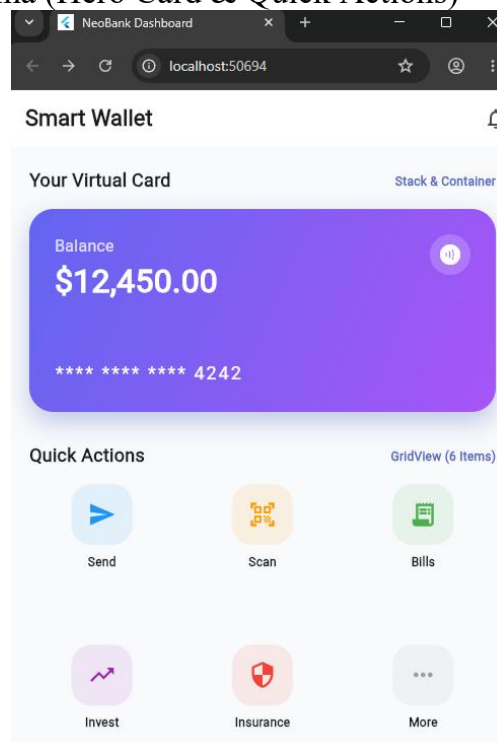
### 3. Penjelasan Code

Kode di atas menerapkan arsitektur antarmuka (UI) berskala produksi dengan mengusung konsep Smart Digital Wallet Dashboard. Kode ditulis secara deklaratif dan modular dengan memanfaatkan fitur keamanan data bawaan Dart (*const compile-time constants*) pada dataset array transaksi untuk mengoptimalkan alokasi memori pada runtime.

Seluruh tata letak halaman dibungkus dalam widget *SingleChildScrollView* terintegrasi guna mengantisipasi kerentanan *UI overflow* (kebocoran layar akibat batasan dimensi piksel). Selain itu, komponen layout statis dikombinasikan secara asinkronus dengan struktur komponen list dinamis, menghasilkan sebuah hierarki halaman dashboard keuangan modern yang adaptif dan berkinerja tinggi.

### 4. Screen Shoot Hasil Running dan Penjelasan Widget

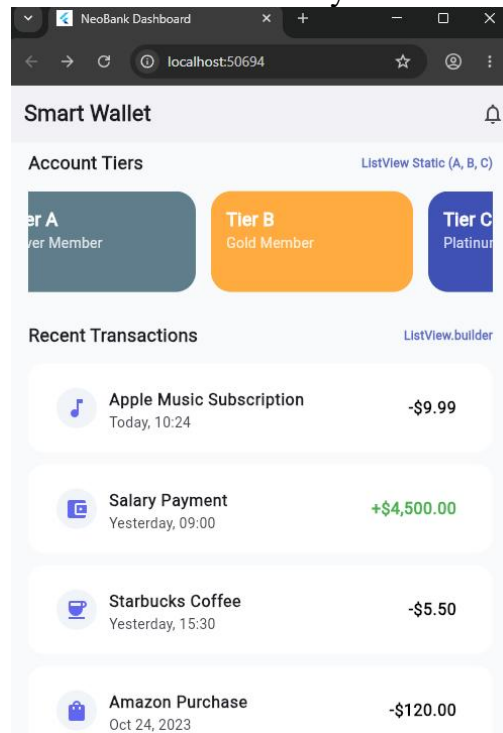
#### A. Tampilan Utama (Hero Card & Quick Actions)



1. Container & Stack: Menampilkan visualisasi Virtual Card premium dengan skema warna gradasi ungu ke merah muda, lengkap dengan tumpukan teks saldo dan nomor kartu.
2. GridView: Menampilkan susunan 6 menu interaktif Quick Actions yang terbagi rapi ke dalam 3 kolom seimbang.



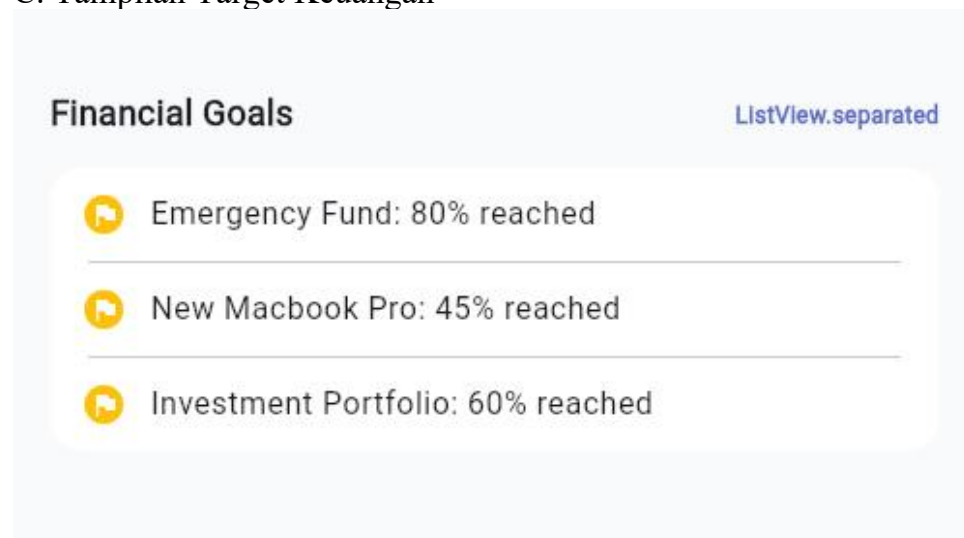
## B. Tampilan Klasifikasi Akun & Riwayat Transaksi



3. ListView (Static): Menampilkan daftar Account Tiers (Tier A, B, C) yang dapat digeser secara horizontal dengan aksen warna yang kontras.

4. ListView.builder: Menampilkan list riwayat transaksi (Recent Transactions) secara dinamis lengkap dengan ikon, deskripsi, serta indikator warna hijau untuk saldo masuk.

## C. Tampilan Target Keuangan



5. ListView.separated: Menampilkan daftar Financial Goals di bagian paling bawah halaman dashboard, di mana setiap item dipisahkan secara otomatis oleh garis pembatas (Divider) abu-abu tipis agar keterbacaan data menjadi lebih optimal.

**~SELESAI~**